

TolC CryoEM Structure in Outer Membrane Vesicles : A Native Cryo-EM Approach

**Florent BEGUIN^{a,b}, Valérie BIOU^a, Bruno MIROUX^a, Martin PICARD^a,
Francesca ZITO^a, Dhenesh PUVANENDRAN^{a,b}**

^a *Institut de Biologie Physico-Chimique, Laboratoire de Biochimie des Protéines Membranaires (UMR7099)
75005, Paris, France*

^b *Université Paris Cité, Paris, France*

Keywords: Extracellular vesicles, Cryo-EM, Structure, Membrane protein, Antibiotics resistance

ABSTRACT

Membrane proteins represent a major challenge in structural and functional biology due to their hydrophobic nature, which complicates both their production and purification. Conventional approaches rely on detergent-based extraction, which can compromise protein stability, and consequently alter native structure and function^{1,2}. To overcome these limitations, we developed a more native approach using outer membrane vesicles (OMVs) directly produced by outer membrane budding of *E. coli* C43, bypassing the need for detergent solubilization entirely.

Our study focuses on AcrAB-TolC, a tripartite efflux pump responsible for multidrug resistance in Gram-negative bacteria and one of the most extensively characterized efflux systems, making it an ideal benchmark for validating new methodologies. As a first step, we characterized TolC in isolation — the outer membrane channel component of the complex responsible for drug efflux across the outer membrane. Cryo-EM analysis of TolC reconstituted within OMVs revealed the protein in both open and closed conformations, a striking observation given that TolC has previously only been resolved in its closed state when purified in detergent and in the absence of its complex partners.

REFERENCES

- 1 T. Cross, et al., *Trends Biochem Sci*, **36**, 117–125, (2011), DOI : [10.1016/j.tibs.2010.07.005](https://doi.org/10.1016/j.tibs.2010.07.005)
- 2 V. Kurauskas, et al., *J Phys Chem Lett*, **9**, 933–938, (2018), DOI : [10.1021/acs.jpclett.8b00269](https://doi.org/10.1021/acs.jpclett.8b00269)